

EARTH OBSERVATION · EVIDENCE-FIRST · AGENT ORCHESTRATED

SatQuery AI

Ask a question about the Earth in plain language.

An LLM agent plans the mission, arbitrates sensors, acquires imagery, runs flood-extent inference, and answers with provenance – or says why it can't.

Why this is hard – and what honesty costs

- ▶ Floods are measured in clouds: monsoon events blind optical satellites,

A value the backend did not supply renders NOT AVAILABLE

with a reason, but never fuses smooth water with radar shadow behind terrain. never a guess, never a demo

fixture in live mode.

- ▶ A natural-language query must become a machine plan: intent

A stage that analysed 40% of

the AOI is marked DEGRADED.

not green.

- ▶ AOI validation, sensor arbitration, acquisition, preprocessing, inference,
- ▶ change detection, evidence assembly – each step auditable.

▶

- ▶ Accuracy is a trap on imbalanced data: water is 10.8% of scorable pixels,

- ▶ so a model predicting 'never water' scores 89.2% accuracy and 0.000 IoU.

SatQuery AI – This is why every figure in this deck is IoU/F1, never

accuracy.

THE HONESTY RULE

Nine services, one gateway, zero browser trust

P1 GATEWAY :8000

Next.js WebRTC proxy,
Browserling gateway only, WS bridge
WebRTC demo ref absolute
BRNEMISBPNQa0000_MODE=1
desertion is a fix to registry
Token is memory-only -
POSTERENTIALS000age;RDPage:6379
logsbgsapthptapeaapb0000 →
acquire → analyse → evidence
BROGMEWH00S/GBAFANA 0tel,
WATIEEERH0000;+8000selectability
Badn00s00At/1600enant.

WEB (live) :3002

SECURITY POSTURE

From prose to evidence: the LangGraph run

- ▶ create_run persists MissionState to Redis agent:run:{job_id}
- ▶ (24 h TTL) behind a process-local cache – any process may
connected ay
- ▶ status_update or poll the run; state is no longer owned by one
status_update or poll the run; state is no longer owned by one
SENSOR_DISAGREEMENT Interpreter.
- ▶ status_update ARBITRATING
- ▶ ACQUIRING_EVIDENCE
- ▶ AGENT_THOUGHT ... Sensor arbitration: cloud forecast blinds optical → SAR
- done (terminal) primary;
- ▶ disagreements are published, not hidden (SENSOR_DISAGREEMENT).
- ▶
- ▶ Every node emits fixed server-side narrative events – user or
model
- ▶ free text never enters the stream; the gateway forwards
verbatim.
- ▶
- ▶ The mission service polls GET /agent/runs/{id} to terminal
state so

LIVE EVENT STREAM (WS)

Measured, not remembered – flood-extent segmentation

- ▶ Dataset: Sen1Floods11 – 400 chips, 10 regions; India + Somalia held out.

baseline 0.204
 flood-unet 0.421
 pretrain 0.407
 hand-only-v2 0.435

F1 0.606 · Pooled IoU (generated report, fingerprint-checked in CI):
 scored (abstained, deterministic Otsu baseline 0.204 flood-unet 0.421
 excluded, not pooled)

- ▶ pretrain 0.407 hand-only-v2 0.435 (pooled F1 0.606)
- ▶
- ▶ Held-out regions generalise (India 0.309 mean IoU vs 0.216 Ghana trained);
- ▶ the report is regenerated by script – never hand-edited.

POOLED IoU

A console that refuses to lie

- ▶ Three-zone shell: query rail | map (the only flexible column) | intel rail;

Exactly one module reads below 1100 px the rails become drawers so the map stays primary.

NEXT_PUBLIC_DEMO_MODE

(enforced by test)

- ▶ QUERY (red) and PARAMETERS (blue) cards mark planner-inferred

Demo: pinned fixtures,

frozen clock, byte-identical, PROGRESS shows only backend-reported stage states – no fake

Playwright-asserts bar.

▶

Live: gateway or nothing –

▶ Map layers: Flood extent, Baseline water, Change, Confidence, AOI; a failed call never falls back

to fixture data. Before/After/Change scrub rail states that 'dates not published'

instead

- ▶ of inventing a calendar.

▶

- ▶ Infrastructure impact reads PostGIS/OSM: roads 2/4 affected (2.46 km),

SatQuery AI – natural language selling intelligence

- ▶ hospitals 1/3 linear extent honestly NOT AVAILABLE for point features

DEMO VS LIVE

The live path this week: shared state in Redis

- ▶ Defect: run state lived in one process – API-created runs were invisible
- ▶ to the poller ('run_not_found'), jobs hung in 'running' forever.
- ▶ Fix: agent:run:{job_id} in Redis (24 h TTL) + process-local cache;
- ▶ mission jobs poll the agent run endpoint to terminal state;
- ▶ dashboard polls /agent/runs/{id} with a bounded 15 s 404 grace window.
- ▶ Proven: mission-container process read the agent's run state from Redis;
- ▶ WS streamed the full envelope chain; job and mission reached honest
- ▶ terminal failed with 'No observations found to analyze.' locally.

TESTS (this branch)

DEMO

The two surfaces

DEMO :3010

When passed a fix address; the
Navigation filter by design runs
by the system while the DEGRADED
status is in effect; re-evaluation
of the available data is required
periodically to determine if the
status can be improved.

Model based system analysis
combines sensor data with the
deterministic baseline, and
the UI says so.

LIVE :3002